

Truth values

The truth value of a mathematical statement is not determined in isolation, but always relative to a universe of discourse or context, in which the statement is interpreted. For example, consider the statement

$$\forall x, y \exists z ((x \neq y) \implies (x < z < y))$$

which may be read as

"For any two distinct numbers x, y , there exists a number z strictly between them."

Note that if the statement is interpreted in the context of natural numbers, it is false because there is no natural number between 1 and 2. But if the statement is interpreted in the context of rational numbers, it is true because for any two rational numbers x, y , the number $z = \frac{x+y}{2}$ is a rational number between them.

- **Definition 1**

Let P and Q be mathematical statements interpreted in some context u , the truth value of $P \wedge Q$ in such a context is fully determined by the following truth table:

P	Q	$P \wedge Q$
1	1	1
1	0	0
0	1	0
0	0	0

- **Remark 1**

The truth table for conjunction can be understood in an even more compact way as follows:

- **Example 1.1**

If $[P]$ is the truth value of P and $[Q]$ is the truth value of Q , then

$$[P \wedge Q] = \min \{[P], [Q]\}$$

This algebraic perspective is useful for replacing structural properties of logical connectives.

- **Definition 2**

Let P and Q be mathematical statements interpreted in some context u , the truth value of $P \vee Q$ in such a context is fully determined by the following truth table:

P	Q	$P \vee Q$
1	1	1
1	0	1
0	1	1
0	0	0

- **Remark 2**

The truth table for disjunction can be understood in an even more compact way as follows:

- **Example 2.1**

If $[P]$ is the truth value of P and $[Q]$ is the truth value of Q , then

$$[P \vee Q] = \max \{[P], [Q]\}$$

- **Definition 3**

Let P and Q be mathematical statements interpreted in some context u , the truth value of $P \Rightarrow Q$ in such a context is fully determined by the following truth table:

P	Q	$P \Rightarrow Q$
1	1	1
1	0	0
0	1	1
0	0	1

- **Remark 3**

In algebraic form, the truth table can be understood in an even more compact way as follows:

- **Example 3.1**

If $[P]$ is the truth value of P and $[Q]$ is the truth value of Q , then

$$[P \Rightarrow Q] = \begin{cases} [Q] & \text{if } [P] = 1 \\ 1 & \text{otherwise} \end{cases}$$

or in another form:

$$[P \Rightarrow Q] = \max \{1 - [P], [Q]\}$$

- **Definition 4**

Let P and Q be mathematical statements interpreted in some context u , the truth value of $P \Leftrightarrow Q$ in such a context is fully determined by the following truth table:

P	Q	$P \Leftrightarrow Q$
1	1	1
1	0	0
0	1	0
0	0	1

- **Remark 4**

In algebraic form, the truth table can be understood in an even more compact way as follows:

- **Example 4.1**

If $[P]$ is the truth value of P and $[Q]$ is the truth value of Q , then

$$[P \Leftrightarrow Q] = \min (1 - [P] + [Q], 1 - [Q] + [P])$$

- **Definition 5**

Let P be a mathematical statement interpreted in some context u , the truth value of $\neg P$ in such a context is fully determined by the following truth table:

P	$\neg P$
1	0
0	1

- **Remark 5**

In algebraic form, the truth table can be understood in an even more compact way as follows:

- **Example 5.1**

If $[P]$ is the truth value of P , then

$$[\neg P] = 1 - [P]$$

We now turn to the question of how truth values are assigned to statements involving quantifiers.

- **Definition 6**

A universally quantified statement of the form $\forall x P(x)$ is true if and only if $P(x)$ is true for all possible values of x . Note that the universally quantified statement $\forall x P(x)$ is false if $P(x)$ is false for some value of x . An existentially quantified statement of the form $\exists x P(x)$ is true if $P(x)$ is true for some value of x . Note that the existentially quantified statement $\exists x P(x)$ is false if $P(x)$ is false for all possible values of x .

- **Remark 6**

In algebraic form, the truth values of the quantifiers $\forall x P(x)$ and $\exists x P(x)$ in a context u are:

- **Example 6.1**

$$[\forall x P(x)] = \min_{x \text{ takes a value in } u} \{[P(x)]\}$$

$$[\exists x P(x)] = \max_{x \text{ takes a value in } u} \{[P(x)]\}$$

- **Definition 7**

We say that two mathematical statements P and Q are logically equivalent, denoted by $P \equiv Q$, if for any context u we obtain $[P] = [Q]$.

- **Example 7.1**

Let P and Q be two mathematical statements, then $\neg(P \wedge Q) \equiv (\neg P \vee \neg Q)$.

P	Q	$P \wedge Q$	$\neg(P \wedge Q)$	$\neg P$	$\neg Q$	$(\neg P \vee \neg Q)$
1	1	1	0	0	0	0
1	0	0	1	0	1	1
0	1	0	1	1	0	1
0	0	0	1	1	1	1

Therefore, $\neg(P \wedge Q) \equiv (\neg P \vee \neg Q)$.

- **Example 7.2**

Let P and Q be two mathematical statements, then $\neg(P \implies Q) \equiv P \wedge \neg Q$.

P	Q	$P \implies Q$	$\neg(P \implies Q)$	$\neg Q$	$(P \wedge \neg Q)$
1	1	1	0	0	0
1	0	0	1	1	1
0	1	1	0	0	0
0	0	1	0	1	0

Therefore, $\neg(P \implies Q) \equiv P \wedge \neg Q$.

- **Example 7.3**

Determine if $\forall x P(x) \equiv \neg \exists x (\neg P(x))$.

- **Solution 7.3.1**

We consider two possible cases for the truth value of $\forall x P(x)$ in any given context u :

Case 1: If $\forall x P(x)$ is true, then $[\forall x P(x)] = 1$. This implies that $[P(x)] = 1$ for all x in u . Consequently, the truth value of its negation is $[\neg P(x)] = 1 - [P(x)] = 0$ for all x in u . Taking the maximum over all x , we get:

$$[\exists x (\neg P(x))] = \max_x \{[\neg P(x)]\} = 0$$

Thus, its negation evaluates to:

$$[\neg \exists x (\neg P(x))] = 1 - 0 = 1$$

Case 2: If $\forall x P(x)$ is false, then $[\forall x P(x)] = 0$. This implies that there exists at least one element x_0 in u such that $[P(x_0)] = 0$. For this specific element, we have $[\neg P(x_0)] = 1 - [P(x_0)] = 1$. Since there is at least one element where $\neg P(x)$ is true, taking the maximum yields:

$$[\exists x (\neg P(x))] = \max_x \{[\neg P(x)]\} = 1$$

Thus, its negation evaluates to:

$$[\neg \exists x (\neg P(x))] = 1 - 1 = 0$$

In both cases, we obtain:

$$[\forall x P(x)] = [\neg \exists x (\neg P(x))]$$

By Definition 2.7, we conclude that $\forall x P(x) \equiv \neg \exists x (\neg P(x))$.

◦ **Example 7.4**

Let P, Q and R be three mathematical statements, we want to prove that the propositions

$$\begin{aligned} A : P \implies (Q \implies R) \\ B : (P \wedge Q) \implies R \end{aligned}$$

are equivalent.

■ **Solution 7.4.1**

We have

P	Q	R	$Q \implies R$	$P \wedge Q$	$P \implies (Q \implies R)$	$(P \wedge Q) \implies R$
1	1	1	1	1	1	1
1	1	0	0	1	0	0
1	0	1	1	0	1	1
1	0	0	1	0	1	1
0	1	1	1	0	1	1
0	1	0	0	0	1	1
0	0	1	1	0	1	1
0	0	0	1	0	1	1

Therefore, $A \equiv B$.

◦ **Example 7.5**

Let P, Q and R be three mathematical statements, we want to prove that the propositions

$$\begin{aligned} A : (P \implies Q) \wedge (Q \implies R) \\ B : (\neg P \vee Q) \wedge (Q \vee R) \end{aligned}$$

are equivalent.

■ **Solution 7.5.1**

Case 1: A is false if and only if

$$(P \implies Q) \text{ is false or } (Q \implies R) \text{ is false}$$

which means

$$\left(\begin{array}{c} P \text{ is true} \\ \text{and} \\ Q \text{ is false} \end{array} \right) \text{ or } \left(\begin{array}{c} Q \text{ is false} \\ \text{and} \\ R \text{ is false} \end{array} \right)$$

Case 2: B is false if and only if

$$(\neg P \vee Q) \text{ is false or } (Q \vee R) \text{ is false}$$

which means

$$\left(\begin{array}{c} P \text{ is true} \\ \text{and} \\ Q \text{ is false} \end{array} \right) \text{ or } \left(\begin{array}{c} Q \text{ is false} \\ \text{and} \\ R \text{ is false} \end{array} \right)$$

In conclusion, both propositions are false exactly in the same cases.

Therefore, $A \equiv B$.

◦ **Example 7.6**

Let P, Q and R be three mathematical statements, we want to prove that the propositions

$$\begin{aligned} A : P &\implies (Q \implies R) \\ B : (P \implies Q) &\implies R \end{aligned}$$

are not equivalent.

■ **Solution 7.6.1**

if P is false, Q is true and R is false, then the proposition A is true and the proposition B is false, then A, B are not equivalent.

◦ **Example 7.7**

Are $A : \neg(P \wedge Q), B : \neg P \wedge \neg Q$ are equivalent?

■ **Solution 7.7.1**

If P is true and Q is false, then A is true and B is false, then A, B are not equivalent.

◦ **Example 7.8**

Are $A : (P \implies Q) \wedge R, B : P \implies (Q \wedge R)$ are equivalent?

■ **Solution 7.8.1**

If P is false, then B is always true. In this case, if R is false, A is false, then A, B are not equivalent.